

DESIGN LAB – OPC Server

Designers:

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Introduction

Our project is an implementation of OPC UA Server utilizing an ESP-32 microcontroller. OPC (Open Platform Communication) defines standards and specifications for industrial telecommunication widely used in devices such as PLC controllers. It specifies communication in real-time between control devices and measurement systems as well as UI controls on capable hardware. There are officially certified open-source implementations available on GitHub certified by OPC Foundation such as open62541 project which our implementation expands upon creating a solution purpose-built to remotely manipulate a Wideband Switch controller made by Łukasz Przystupa. During the design process two versions of a custom OPC Client were created to provide better control and handling of this particular task than general purpose clients available. Apart from OPC protocol the device also supports direct UART bridge to the Raspberry Pi Pico used inside of the switch.

Available UART commands mimic the ones provided by switch controller:

- in <1-2> out <1-16> - attach output to an input
- in <1-2> off - turn off input
- release - release controller
- clear - Clear display
- draw - Redraw display
- screen - Enable screensaver
- status - Get current status

Examples:

- in 1 out 5 - Set IN1 to OUT5
- in 2 out 12 - Set IN2 to OUT12
- in 1 off - Turn off IN1

Project timeline:

- 30.10.25 - initial commits and testing
- 06.11.25 - Added basic UART functionalities and needed OPC nodes
- 13.11.25 - Modified UART to be compliant with pico commands
- 20.11.25 - Added status command to Pico, refined UART and OPC node code on ESP
- 27.11.25 - Final touches in regards to main functionality
- <4,11,18>.12.25 - Extensive stability testing and bug fixes
- <8,15>.01.26 - Python OPC client TUI / GUI development
- 22.01.26 - Minor bug fixes
- 29.01.26 - Final presentation